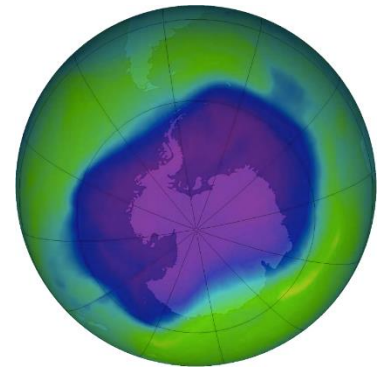


Ozone Layer Depletion

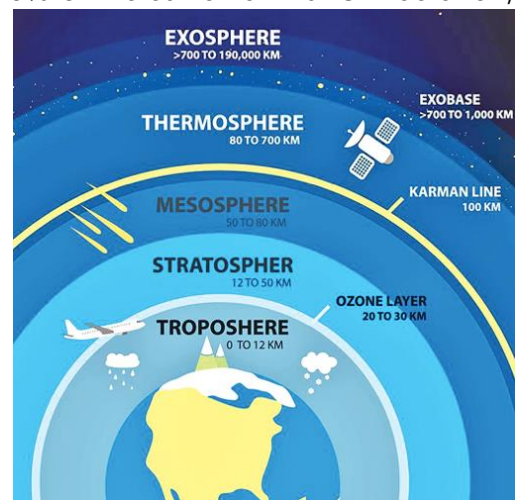
What is the ozone layer?

The ozone layer is a region of Earth's stratosphere that absorbs most of the Sun's ultraviolet radiation. This layer contains a high concentration of ozone (O_3) as compared to other parts of the atmosphere. The concentration of ozone in this layer is < 10 parts per million of ozone, whereas the average ozone concentration in Earth's atmosphere as a whole, is 0.3 parts per million. This layer is located mainly in the lower portion of the stratosphere, ranging from approximately 15-35 km above Earth. And its thickness varies seasonally and geographically.



Role of ozone in the stratosphere

Ozone is present in the lower stratosphere. It absorbs ~95% of the sun's harmful UV radiation, thus preventing it from reaching the earth's surface. UV radiation is classified based on its wavelength into UV-A (400-315 nm), UV-B (315-280 nm), and UV-C (280-100 nm). Amongst these, UV-C, which is very harmful to all living things, is entirely screened out by ozone at around 35 kilometres altitude. UV-B radiation harmful is the main cause of sunburn, it can also cause cataracts and problems such as skin cancer. The ozone layer is very effective at screening out most of the UV-B as well, even though some longer wavelengths do reach the surface. Most of the UV-A, which is significantly less harmful, is not absorbed by ozone, and reaches the earth's surface.



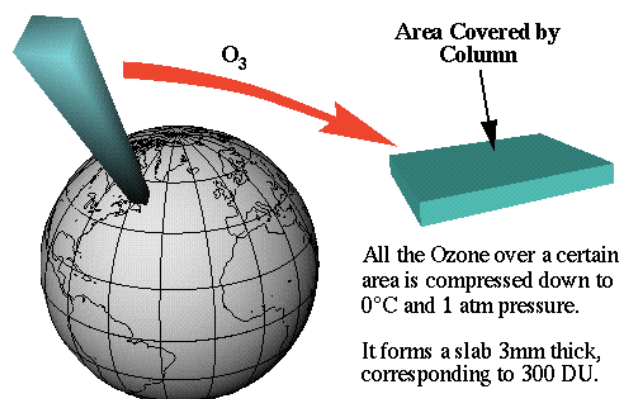
Distribution in the stratosphere

The thickness of the ozone layer varies worldwide, and is generally thinner near the equator and thicker near the poles. Thickness refers to how much ozone is present in a column over a given area, and varies from season to season. The reasons for these variations are mostly due to atmospheric circulation patterns and solar intensity.

Ozone hole and its measurement

The ozone hole is not technically a "hole" where ozone is completely absent, but rather, it is a region where ozone is exceptionally depleted in the stratosphere over the Antarctic circle. The thickness of the ozone layer is measured across the entire atmospheric column from the surface to the edge of space. If the total amount of ozone was compressed to a pressure of 1 atm, its height in mm is given by Dobson units. 1 mm: 100 Dobson (DB) units; 2 mm: 200 Dobson units and so on. The global average thickness of ozone is 300 DB, and in the ozone 'hole', the thickness

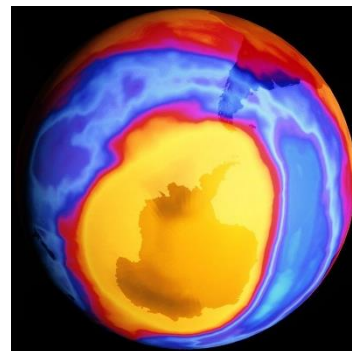
Dobson Unit



reduces to 100 DB. The ozone hole is not permanent, it thins down and thickens up at times. Ozone values less than 220 DB are termed as Ozone holes.

Certain observations about the ozone hole

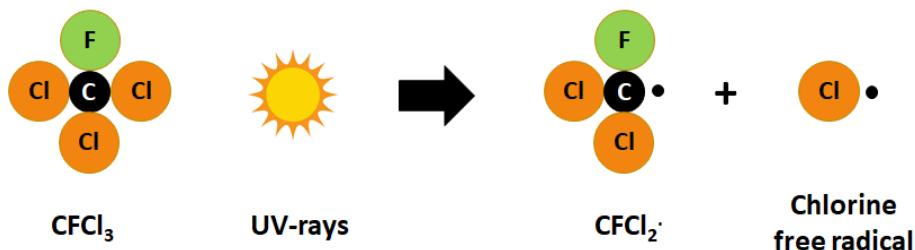
The size of the ozone hole in the Antarctic covers ~20-23 million km² depending on the seasons. The largest, single-day ozone hole was recorded by satellite to be 29.9 million km² in Sept. of 2000. The southern polar ozone hole usually lasts from mid Aug to end of Dec., while the northern polar hole, over the Arctic circle, lasts a few days. So, the Ozone hole is largest above the Antarctic, and just to put its size in perspective, its area is three times larger than the entire land mass of the USA.



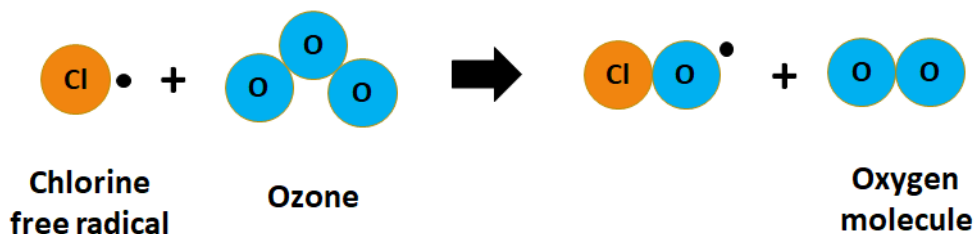
What causes depletion of ozone?

The main causes of the ozone hole are certain man-made chemicals containing halogens, that are used as refrigerants, solvents, propellants, and foam-blowing agents (chlorofluorocarbons (CFCs), HCFCs, halons), referred to broadly as ozone-depleting substances (ODS). So, how exactly do these chemicals reach the stratosphere and cause ozone depletion? CFC molecules are made up of chlorine, fluorine and carbon atoms and are extremely stable. Most molecules decompose before they can cross into the stratosphere from the troposphere. CFC's, being extreme stable, make their way into the stratosphere, reaching very high altitudes, where photons are more energetic. When the CFC's come into contact with these high energy photons, their individual components are freed from the whole.

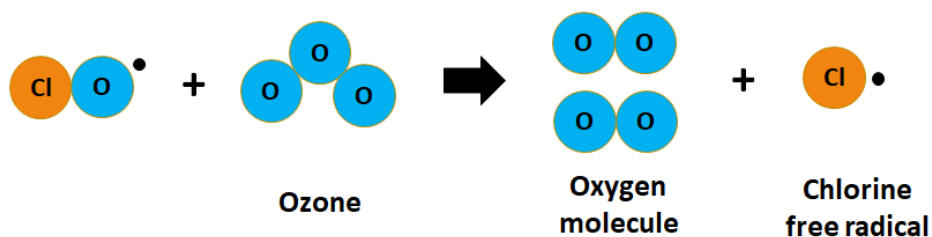
- When UV lights hits a halocarbon, such as chlorofluorocarbon such as CFCl_3 , one chlorine atom breaks off, leaving CFCl_2 radical and Cl radical.



- The chlorine free radical attacks an ozone molecule, pulling out an oxygen atom, leaving behind an O_2 molecule.



- The Cl and O atoms combine to form a chlorine monoxide molecule (ClO). X

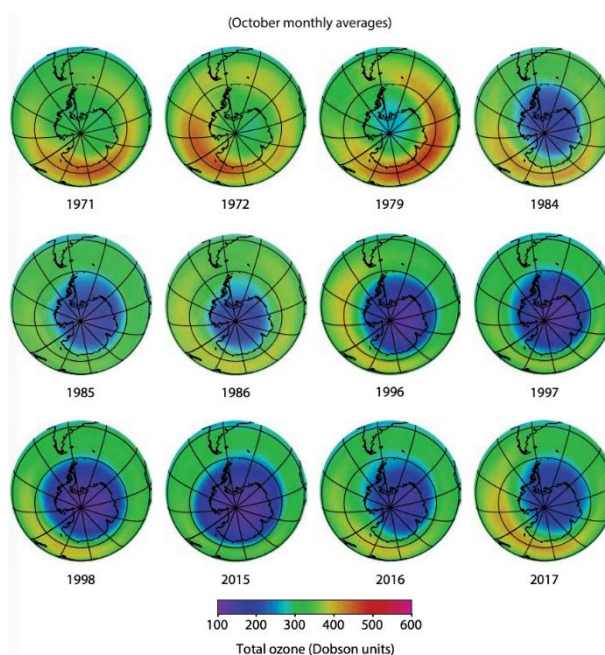


- Another oxygen radical pulls out the oxygen from ClO, leaving the Cl free radical free to attack another ozone molecule and begin the cycle all over again. So, Cl acts as a catalyst here. One chlorine atom can thereby destroy thousands of ozone molecules.

Reasons for the massive ozone hole over Antarctica

So, why is the ozone hole so massive in size over Antarctica, and not in other regions? People don't even live there and emit harmful gases. So, let's look at it from a scientific and mechanistic point of view. The ozone hole occurs during Antarctic springs, i.e., from Sept. to early Dec., during this time, ~50% of ozone is destroyed. Let us try and understand the reason for its seasonal depletion.

- Antarctica has the coldest winters on earth, often reaching -80°C . These extreme temperatures result in the formation of polar stratospheric clouds (PSC's) which are basically, a conglomeration of frozen H_2O and HNO_3 . Due to their extremely cold temperatures, PSC's form an electrostatic attraction with CFC molecules as well as other halogenated compounds.
- During winter, there is no sunlight to drive chemical reactions. The frozen crystals, of water and nitric acid, that make up polar stratospheric clouds, provide a surface for the reactions that will free chlorine atoms in the Antarctic stratosphere. These reactions are initiated in spring, with the arrival of the sun, and high energy photons, causing the PSCs to melt.
- As a result, CFCs react with the high energy photons, releasing Cl radical, which then causes the depletion of the ozone layer.
- So, to summarize, extreme low temperature and solar energy, both of which are readily available in the Antarctic spring, result in a large-scale ozone layer depletion.



So, how can we combat this depletion in the ozone layer?

The role of CFCs in depleting the ozone layer was identified by two scientists, Molina and Rowland, from the USA, who were later awarded the Nobel prize in 1995 for their work on this problem. Countries across the globe began recognizing the seriousness of the issue, which resulted in the Montreal protocol. The Montreal Protocol is an international treaty that was designed to protect the ozone layer by phasing out the production substances responsible for

ozone depletion. It began with the Vienna convention for the Protection of the Ozone Layer, which was signed on 22nd March 1985, and was followed by the Montreal Protocol on 16th September 1987. It was universally ratified by all 197 nations of the United Nations, and came into force on 16 September 1989.

On the 19th June 1991, India became a party to the Vienna convention, and on the 17th September 1992, India became a party to the Montreal Protocol. As a result of these treaties, the ozone hole in Antarctica is slowly, but steadily recovering. Climate projections indicate that the ozone layer will return to 1980 levels between the years 2050 and 2070. The Montreal Protocol is a landmark agreement that has successfully reduced the global production, consumption and emission of ozone-depleting substances (ODSs), which are also greenhouse gases that contribute to global warming. Under this treaty, all countries have accepted legally binding obligations to phase out the production and consumption of ODSs including CFCs, halons and other substances that release chlorine or bromine into the atmosphere.

Ozone depleting substances (ODSs)

Ozone depleting substances were defined under the Montreal Protocol, as chemicals that destroy the ozone layer. Several groups of halogenated hydrocarbons (~96) were identified under this category. They contain either chlorine or bromine (fluorine- only-containing compounds do not harm the ozone layer). ODSs are categorized into two groups, namely, the Class I ODSs, such as chlorofluorocarbons (CFCs), and Class II ODS, such as hydrochlorofluorocarbons (HCFCs). Chlorofluorocarbons (CFCs): are a group of compounds containing C, F, Cl, whereas Hydrochlorofluorocarbons (HCFCs): group of compounds containing H, C, F, Cl. Class II ODSs include Hydrofluorocarbons (HFCs): which were produced mostly in developed countries. Since they contain only, H and F, they do not harm the ozone layer, but they have been identified as GHG with a high global warming potential (GWP) as compared to CFCs and HCFCs. Halomethanes, Carbon tetrachloride (CCl₄) and Hydrogen chloride (HCl) have also been classified as ODSs by the Montreal Protocol.

Uses of ODSs in everyday life

1. CFCs used as propellants in Inhalers
2. Fire extinguishers: contain haloalkanes
3. Hairsprays: originally contained CFCs, now HFCs
4. CFCs and HCFCs are used as propellants in Pest sprays
5. Foam insulation contains HCFCs.

The year 2012 marked the 25th anniversary of the signing of the Montreal Protocol. Since its inception, the world has phased-out 98% of the Ozone-Depleting Substances (ODS) contained in nearly 100 hazardous chemicals worldwide; every country is in compliance with stringent obligations; the MP has achieved the status of the first global regime with universal ratification.